

Enhanced Power Generation From PV Array Under Partial Shading Conditions by Shade Dispersion Using Su Do Ku Configuration

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Abstract—Partial shading of PV arrays reduces the energy yield of PV systems and the arrays exhibit multiple peaks in the P-V characteristics. The losses due to partial shading are not proportional to the shaded area but depend on the shading pattern, array configuration and the physical location of shaded modules in the array. This paper presents a technique to configure the modules in the array so as to enhance the generated power from the array under partial shading conditions. In this approach, the physical location of the modules in a Total Cross Tied (TCT) connected PV array are arranged based on the Su Do Ku puzzle pattern so as to distribute the shading effect over the entire array. Further, this arrangement of modules is done without altering the electrical connection of the modules in the array. The Su Do Ku arrangement reduces the effect of shading of modules in any row thereby enhancing the generated PV power. The performance of the system is investigated for different shading patterns and the results show that positioning the modules of the array according to “Su Do Ku” puzzle pattern yields improved performance under partially shaded conditions.

Index Terms—Array configuration, mismatch, partial shading, photovoltaic.

I. INTRODUCTION

THE depletion of conventional energy resources and the growing concern over environmental problems has increased the contribution of renewable energy sources to the total energy consumption [1], [2]. Solar energy is acquiring much acclaim due to the decreasing cost of photovoltaic cells and recent advancements in power conversion technology [3]. Besides being maintenance free, the solar power is also eco friendly [4], [5]. The PV application ranges from small installations of a few kW located in individual premises to large power plants generating several MW of power [6], [7].

Generally, the panels are connected in series and parallel to meet the load power requirement. The output power of the PV array decreases considerably when one or more of the panels in the array are subjected to shading [8], [9]. Partial shading can be caused by the shadows of buildings, trees and poles. In large PV systems, moving clouds also leads to partial shading [10], [11]. Partial shading complicates the PV characteristics with multiple peaks [12], [13] apart from reducing the energy

yield of PV systems. The occurrence of such multiple peaks can mislead some MPPT algorithms to get trapped at local peaks [14], [15]. Different techniques have been proposed [16], [17] to track the global peak under partial shading conditions. The issue of predicting the maximum power under these conditions is reported [18].

The losses due to partial shading are not proportional to the shaded area but depend on the shading pattern, array configuration and the location of shaded module in the array. Different array configurations have been proposed in literature [19], [20] to reduce the mismatch losses in the array. Three interconnection schemes viz., Series-Parallel (SP), Total Cross Tied (TCT) and Bridge Linked (BL) are compared for their losses, maximum power, fill factor, reliability and energy yield due to mismatch caused by the manufacturer’s tolerances in cell characteristics and by partial shading [21]. A study of the operational life-time of PV arrays based on the probability theory indicates that the introduction of cross ties (TCT or BL schemes) in the array almost doubles the life time of the array [22]. A method [23] has been proposed to predict the maximum power production from the existing PV systems. Changing the interconnection schemes of the modules from SP to TCT increases the power by more than 4% [24] and the TCT configuration is considered as the best solution to mitigate the mismatch losses under partially shaded conditions. Comparison in terms of maximum power and fill factor has been carried out [25] on SP, TCT, BL, Simple Series (SS) and Honey Comb (HC) configurations. The TCT configuration ranks higher compared to other schemes in terms of maximum power. An Electrical Array Reconfiguration (EAR) controller [26] is proposed to optimize the performance of PV powered volumetric pump under different insolation levels. Based on the irradiance level, the EAR controller chooses a suitable configuration for different operating conditions of the pump. In adaptive reconfiguration scheme [27], a switching matrix connects a solar adaptive bank to a fixed part of the array so that less shaded modules of the adaptive part are in parallel with more shaded rows of fixed part. A large number of switches and sensors are required to implement the technique. An electrical reconfiguration technique [28] is proposed in which the connections of the modules are dynamically changed according to a switching matrix so as to maximize the current of the single string in the event of shading. Though the power loss is reduced under several partial shaded conditions, the technique involves high cost and greater complexity. The reconfiguration strategy [29] reported in literature ensures a minimum deviation from the PV operating voltage and the strategy is based on removing the

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shaded modules from the array by sensing the current through the bypass diode. A switching network is required to connect and disconnect the modules. Moreover, individual current sensors are required to sense the current through each bypass diode.

This paper presents a method to configure the physical placement of the modules in a TCT connected PV array to enhance the PV power generation under partial shaded conditions. The modules are arranged based on the Su Do Ku puzzle pattern without altering their electrical connection within the array. This structure facilitates to distribute the effect of shading over the array thereby reducing the occurrence of shading of modules in the same row. The performance of the system is investigated for different shading patterns and the results reveal that the Su Do Ku configured structure exhibits superior performance under partially shaded conditions.

The rest of the paper is arranged as follows: Section II describes the modeling and details of TCT PV array considered for the study. Further the proposed Su Do Ku configuration and the consequent shading implications on the array under different patterns of partial shading are discussed. The results and discussion are presented in Section III. Section IV lists the conclusions from the study.

II. SYSTEM DESCRIPTION

A. Model of a PV Cell

A single PV module consists of a number of PV cells in series and a PV cell is modeled as a current source shunted with a diode and represented by an equivalent circuit [30] shown in Fig. 1(a). The equation relating the output current and the voltage of a PV module at an insolation G can be written as

$$I_m = I_{ph} - I_o \left[\exp \left(\frac{V_{pv} + R_s I_m}{A} \right) - 1 \right] - \left[\frac{V_{pv} + R_s I_m}{R_{sh}} \right] \quad (1)$$

where I_m is the current generated by the module, I_{ph} is the photoelectric current, I_o is the saturation current, V_{pv} is the PV voltage and $A = nkT/q$. T represents the temperature of the module in Kelvin, k is the Boltzmann's constant, q is the electric charge, n is the number of cells in series, R_s and R_{sh} are the series and shunt resistance respectively. The light generated current [30] is obtained as the function of short circuit current as

$$I_{ph} = I_{sco} \left(\frac{G}{G_o} \right) (1 + \alpha_1 (T - T_o)) \frac{(R_s + R_{sh})}{R_{sh}} \quad (2)$$

where I_{sco} , is the short circuit current of the module at standard insolation G_o (1000 W/m^2) and standard temperature T_o (25°C) and α_1 is the module's temperature coefficient for current. The above equations are used to model the PV modules considered for study. The specifications of the module at standard test conditions are shown in Table I.

B. TCT Configured PV Array

A 9×9 PV array connected in TCT configuration Fig. 1(b) is considered for the study. A TCT configuration is obtained from the simple series parallel configuration by connecting cross ties across each row of the junctions. All the modules in a column

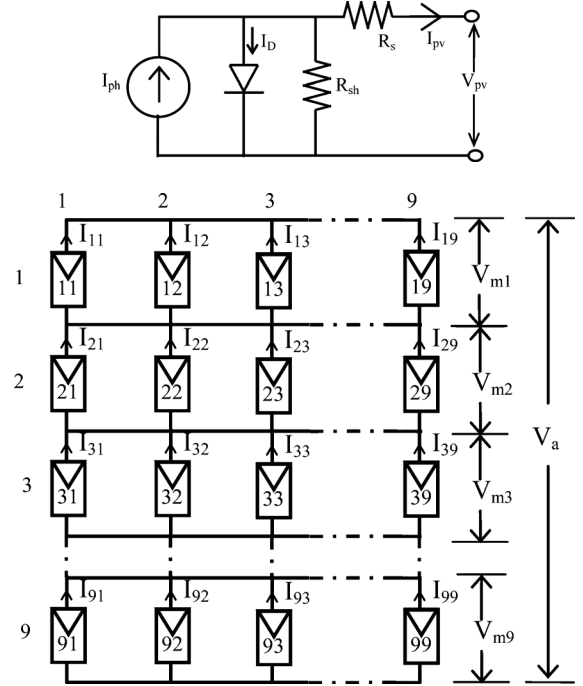


Fig. 1. (a) Equivalent circuit of a PV cell. (b) PV array connected in TCT configuration.

TABLE I
PV SPECIFICATIONS AT 1000 W/m^2 , 25°C

PV power	80 W
Open circuit Voltage	22 V
Short circuit current	4.7 A
Nominal Voltage	18 V
Nominal Current	4.4 A

are connected in series and each module in a row has modules connected in parallel. The PV array consists of 81 modules with nine rows and nine columns. The modules are labeled as 'pq' where p denotes the row and q, the column in which the module is connected. For example the module numbered 24 represents the module situated in the second row and fourth column. The current generated by a module at an irradiance G is given by

$$I = k I_m \quad (3)$$

where I_m is the current generated by the module at standard irradiance $G_o = 1000 \text{ W/m}^2$ and $k = G/G_o$. Therefore the current generated by the module is directly proportional to the solar irradiance on the panel. The voltage of the array is given by the sum of the voltages of the 9 rows. Applying the KVL

$$V_a = \sum_{p=1}^9 V_{mp} \quad (4)$$

where V_a is the voltage of the PV array and V_{mp} refers to the voltage of the panels at the p^{th} row.

Using the Kirchhoff's current law, the current at each node can be expressed as

$$I_a = \sum_{q=1}^9 (I_{pq} - I_{(p+1)q}) = 0, \quad p = 1, 2, 3, \dots, 8. \quad (5)$$

11	42	53	94	25	76	87	68	39
21	92	73	84	35	66	57	18	49
31	82	63	44	55	16	97	78	29
41	32	13	54	85	96	77	28	69
51	22	93	64	75	46	17	38	89
61	72	83	24	15	36	47	98	59
71	12	23	34	45	56	67	88	99
81	62	43	74	95	26	37	58	19
91	52	33	14	65	86	27	48	79

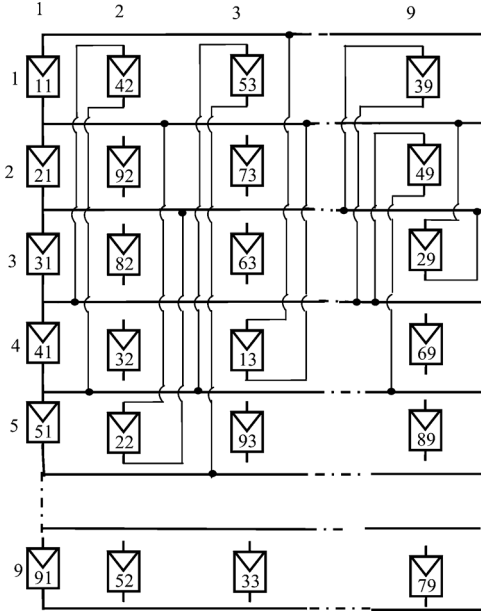


Fig. 2. (a) Chosen Su Do Ku pattern. (b) Su Do Ku arrangement.

C. Su Do Ku Configured Structure for PV Array

Su Do Ku is a logic based number placement puzzle. The Su Do Ku puzzle pattern involves nine 3×3 matrices and every row, column, and matrix accommodates the digits 1 to 9 without repeating any number. A Su Do Ku puzzle pattern shown in Fig. 2 is chosen for the study. The first digit in each box represents the number and the second digit denotes the column of the 9×9 array. It can be seen that each sub matrix, each row and each column of the array accommodates all the digits 1 to 9. To show that Su Do Ku configuration will enhance the PV power generation, the modules in the TCT arrangement [Fig. 1(b)] are rearranged according to the chosen puzzle pattern and is shown in Fig. 2(b). Here the electrical connection of the module remains the same as indicated in Fig. 1(b). However, the physical location of the modules is changed. The panel 42 (fourth row, second column) is physically moved to the first row second column but the panel connection remains in the fourth row. Similarly, the panel 22 which belongs to second row second column is moved to fifth row in the same column. However, the panel remains connected to second row. In this way, the physical locations of the panels are changed without altering the electrical connections in the array.

Since the electrical configuration of the array does not change, the voltage and current equations remain the same as in the TCT arrangement. The panels belonging to the same

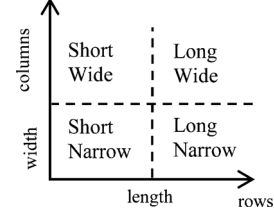


Fig. 3. Different shading conditions.

row in the TCT arrangement are moved to different rows in the Su Do Ku arrangement. This enables to reduce the shading of the panels in the same row and distribute the shading over the entire array. Thus, the Su Do Ku arrangement increases the current entering a node under partial shading conditions and minimizes the bypassing of panels. Hence the power generated by the array for the same shading pattern is increased.

The electrical reconfiguration technique reported in [28] also aims to distribute the effect of shading. However, the main difference between the reconfiguration technique and the proposed technique is that in reconfiguration method, the electrical connections of the modules are dynamically changed based on the shading conditions whereas in the proposed technique it remains fixed under all shading conditions. The practical implementation of the reconfiguration technique involves a number of voltage and current sensors in addition to the switches required to change the electrical connections of the modules. As the modules are configured permanently, the proposed “Su Do Ku”-based method does not require any sensors or switches. Moreover no separate control algorithm is required as in case of reconfiguration technique.

In large PV systems, with reconfiguration technique, the number of sensors and switches increases with array size and control circuit becomes complex. With the proposed system, the entire PV array can be divided into a 9×9 sub assembly and the proposed technique can be employed to ensure that the modules in the row are kept apart. Further, this arrangement of modules as per the Su Do Ku pattern needs to be made just once only since the same pattern holds effective for any shading pattern.

To verify the proposed technique, four different shading conditions namely Short Wide (SW), Short Narrow (SN), Long Wide (LW), and Long Narrow (LN) are considered. The shading conditions are defined [31] based on the number of shaded columns (width) and shaded modules per column (length) (Fig. 3). The PV characteristics are obtained for each shading condition for both the TCT and the Su Do Ku arrangements and the results are presented in Section III.

III. RESULTS AND DISCUSSION

To evaluate the performance of the proposed method, a 9×9 PV array connected in TCT configuration is subjected to four different shading patterns. The location of GP in both the TCT and the Su Do Ku arrangement are calculated theoretically. The location of GP indicates the number of rows that are bypassed to extract the maximum power. The theoretical results are verified using simulation studies in MATLAB/Simulink environment. The PV characteristics are obtained for both the TCT and the

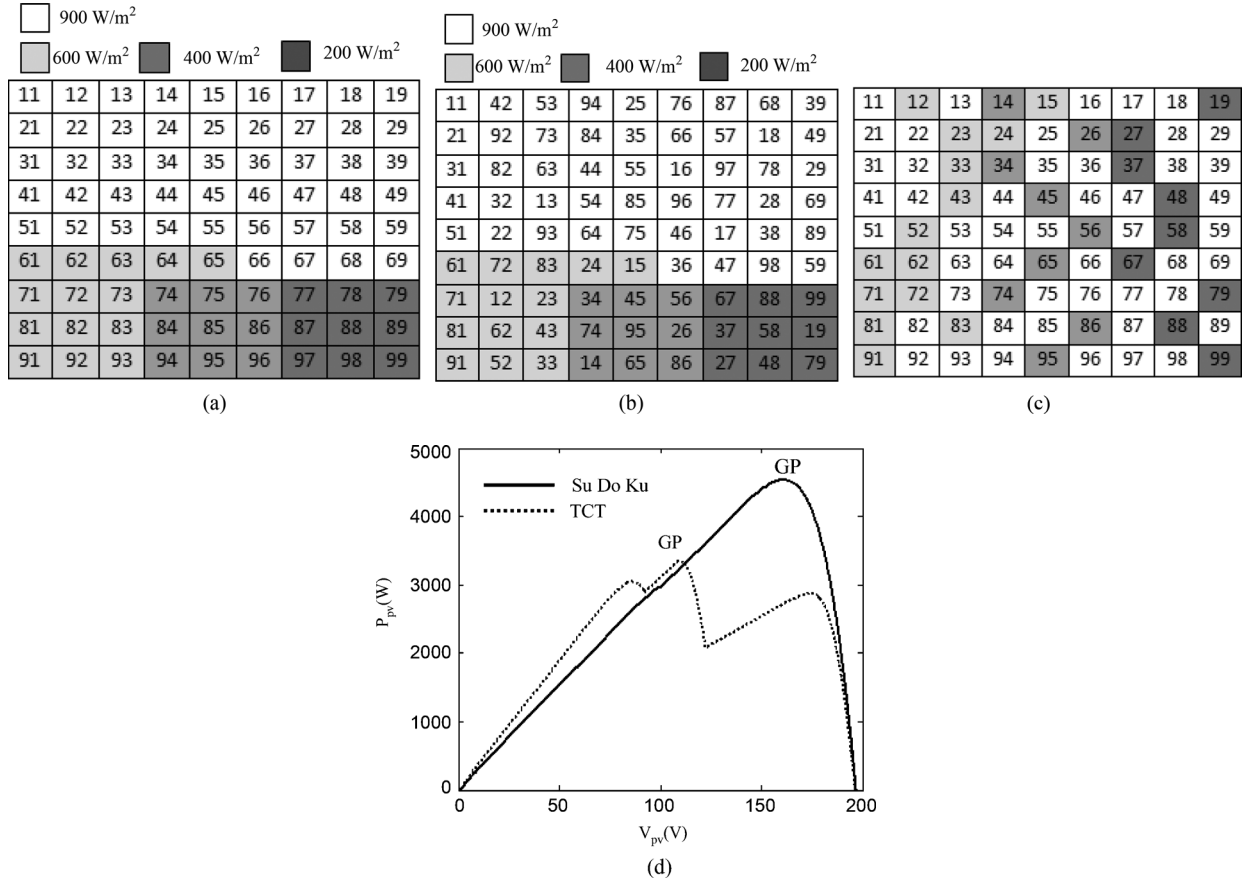


Fig. 4. Shading pattern for case 1. (a) TCT arrangement. (b) Su Do Ku arrangement. (c) shade dispersion with Su Do Ku arrangement. (d) PV characteristics of the array for TCT and Su Do Ku arrangement for case 1.

Su Do Ku arrangement for each shading pattern and the results are presented.

Case 1 (Short Wide):

A PV array with four distinct groups is considered. The group one receives an insolation of 900 W/m^2 while group two, group three, and group four receives 600 W/m^2 , 400 W/m^2 , and 200 W/m^2 respectively and the shading pattern is described in Fig. 4(a). Since all the columns and less number of rows are subjected to shading, the shading condition is referred to as short wide. To find the location of GP, it is necessary to calculate the current generated in each row in the array. Here all the panels in row 1 receive a uniform insolation of 900 W/m^2 and the current in row 1 is expressed as

$$I_{R1} = k_{11}I_{11} + k_{12}I_{12} + k_{13}I_{13} + k_{14}I_{14} + k_{15}I_{15} + k_{16}I_{16} + k_{17}I_{17} + k_{18}I_{18} + k_{19}I_{19} \quad (6)$$

where $k_{11} = G_{11}/G_o = 0.8$ where G_{11} is the solar irradiance of the panel numbered 11 and I_{11} is the current generated by the panel. Assuming the current generated by all the panels under STC as I_m

$$I_{R1} = 9 \times 0.9I_m. \quad (7)$$

All the panels in row 2 to 5 also have uniform insolation of 900 W/m^2 . Therefore

$$I_{R1} = I_{R2} = I_{R3} = I_{R4} = I_{R5}. \quad (8)$$

In row 6, four panels receive 800 W/m^2 while the remaining five panels are subjected to 600 W/m^2 . The current generated by the panels in other rows are expressed as

$$\begin{aligned} I_{R6} &= 5 \times 0.6I_m + 4 \times 0.9I_m \\ I_{R7} &= I_{R8} = I_{R9} = 3 \times 0.6I_m + 3 \times 0.4I_m \\ &\quad + 3 \times 0.2I_m. \end{aligned} \quad (9)$$

Since the current generated in different rows is different, there exist multiple peaks on the PV characteristics. Now to identify the location of GP, the module currents are noted in Table II according to the order in which panels will be bypassed.

Neglecting the small variations in voltage across each row, the voltage of the array $V_a = 9 V_m$, if none of the panels are bypassed and $V_a = 8 V_m + V_d$, where V_d is the voltage across the diode if a single row is bypassed. As $V_d \ll V_a$, V_d is neglected.

The power produced by the array

$$P_a = I_m V_a = I_m(9V_m) \quad (10)$$

if no array is bypassed. The voltage of the array and the corresponding power are calculated and are noted in Table II. It is observed that only when rows 6, 7, 8, and 9 are bypassed, maximum power is extracted from the array, i.e., the location of GP is far away from the nominal voltage of the array. To verify theoretical results, simulations are carried out in MATLAB/Simulink environment and the PV characteristics

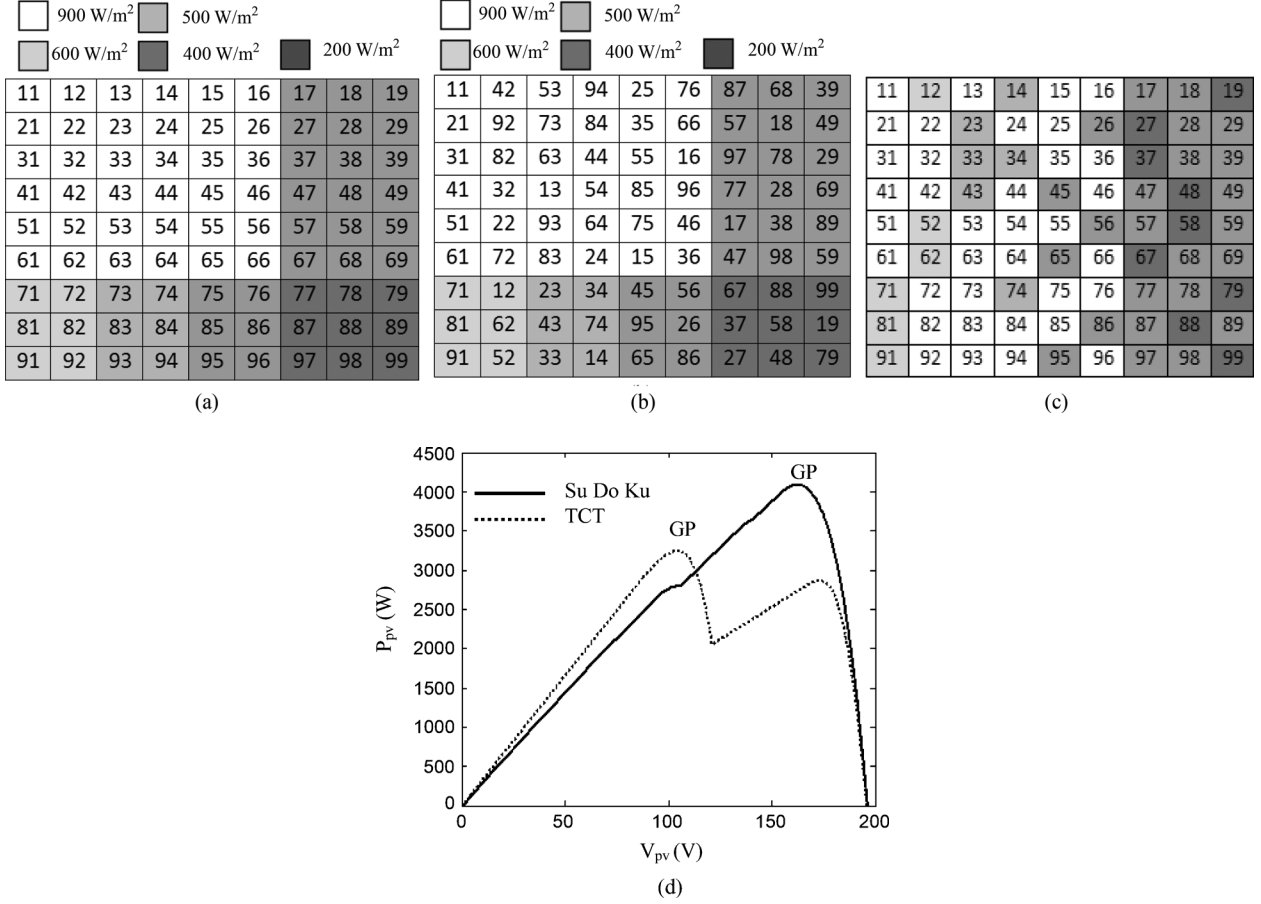


Fig. 5. Shading pattern for case 2. (a) TCT arrangement. (b) Su Do Ku arrangement. (c) Shade dispersion with Su Do Ku arrangement. (d) PV characteristics of the array for TCT and Su Do Ku arrangement for case 2.

corresponding to this array is plotted in Fig. 4(d). It can be observed that the maximum power generated by the array is 3348 W and the GP occurs at 109.4 V which is less than the nominal voltage of the array.

To validate the proposed arrangement of modules according to Su Do Ku puzzle pattern, the rearranged array is subjected to the same shading pattern [Fig. 4(b)] and the current in each row is calculated as

$$\begin{aligned}
 I_{R1} &= I_{R2} = 5 \times 0.9I_m + 2 \times 0.6I_m + 0.4I_m + 0.2I_m \\
 I_{R3} &= I_{R4} = I_{R5} = 6 \times 0.9I_m + 0.6I_m + 0.4I_m \\
 &\quad + 0.2I_m \\
 I_{R6} &= I_{R7} = I_{R8} = 5 \times 0.9I_m + 2 \times 0.6I_m \\
 &\quad + 0.4I_m + 0.2I_m \\
 I_{R9} &= 6 \times 0.9I_m + 0.6I_m + 0.4I_m + 0.2I_m. \quad (11)
 \end{aligned}$$

The module currents and voltage are noted in Table II according to the order in which panels will be bypassed. From Table II, it is clear that after the rearrangement of modules according to Su Do Ku puzzle pattern, the location of GP shifts to the nominal voltage of the array. Moreover, the power generated by the array is increased from 3348 to 4532 W which is 26.12% higher than the TCT arrangement and the PV characteristic of the array is plotted in Fig. 4(d). The increase in power is due to the relocation of modules which distributes the effect of shading over the entire array [Fig. 4(c)].

TABLE II
LOCATION OF GP IN TCT AND SU DO KU ARRANGEMENT -CASE 1

TCT arrangement			Su Do Ku puzzle pattern		
Row current in the order in which panels are bypassed	voltage V_a	power P_a	Row current in the order in which panels are bypassed	voltage V_a	power P_a
I_{R9}	$3.6I_m$	$9V_m$	$32.4 V_m I_m$	I_{R6}	$6.3 I_m$
I_{R8}	$3.6I_m$	-	-	I_{R7}	$6.3 I_m$
I_{R7}	$3.6I_m$	-	-	I_{R8}	$6.3 I_m$
I_{R6}	$6.6I_m$	$6V_m$	$39.6 V_m I_m$	I_{R1}	$6.3 I_m$
I_{R5}	$8.1I_m$	$5V_m$	$40.5 V_m I_m$	I_{R2}	$6.3 I_m$
I_{R4}	$8.1I_m$	-	-	I_{R4}	$6.6 I_m$
I_{R2}	$8.1I_m$	-	-	I_{R3}	$6.6 I_m$
I_{R3}	$8.1I_m$	-	-	I_{R5}	$6.6 I_m$
I_{R1}	$8.1I_m$	-	-	I_{R9}	$6.6 I_m$

Case 2 (Long Wide):

A PV array with five distinct groups according to the insolation level is considered. The group one receives an insolation of 900 W/m^2 , while group two, group three, group four, and group five receives 600 W/m^2 , 500 W/m^2 , 400 W/m^2 , and 200 W/m^2 respectively and the shading pattern is shown in Fig. 5(a). The current in each row is calculated as

$$\begin{aligned}
 I_{R1} &= 6 \times 0.9I_m + 3 \times 0.4I_m \\
 I_{R2} &= I_{R3} = I_{R4} = I_{R5} = I_{R6} = I_{R1} \\
 I_{R7} &= I_{R8} = I_{R9} = 2 \times 0.9I_m + 2 \times 0.5I_m + 2 \\
 &\quad \times 0.4I_m + 3 \times 0.2I_m. \quad (12)
 \end{aligned}$$

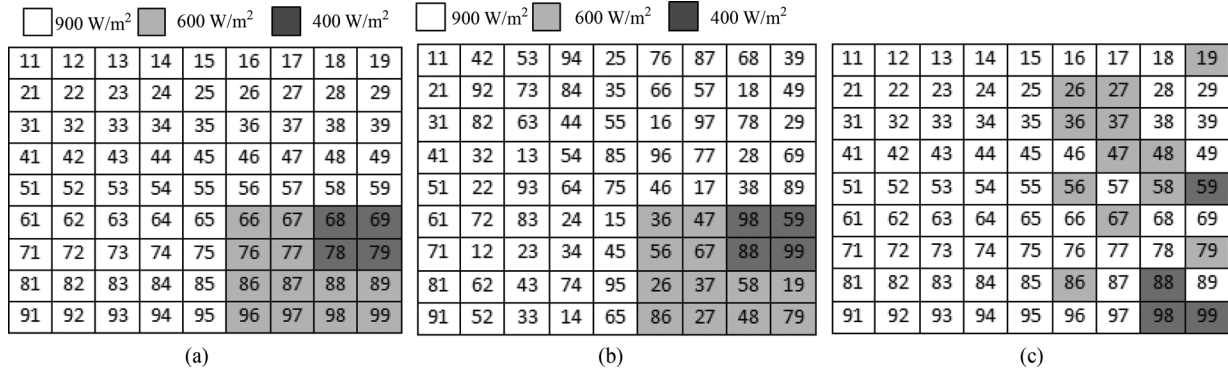


Fig. 6. Shading pattern for case 3. (a) TCT arrangement. (b) Su Do Ku arrangement. (c) Shade dispersion with Su Do Ku arrangement. (d) PV characteristics of the array for TCT and Su Do Ku arrangement for case 3.

TABLE III
LOCATION OF GP IN TCT AND SU DO KU ARRANGEMENT -CASE 2

TCT arrangement			Su Do Ku puzzle pattern		
Row current in the order in which panels are bypassed	voltage V_a	power P_a	Row current in the order in which panels are bypassed	voltage V_a	power P_a
Ir7	3.6 I _m	9V _m	Ir6	5.4 I _m	9 V _m
Ir8	3.6 I _m	-	Ir2	5.5 I _m	8 V _m
Ir9	3.6 I _m	-	Ir4	5.5 I _m	-
Ir1	6.6 I _m	6 V _m	Ir3	5.6 I _m	6 V _m
Ir2	6.6 I _m	-	Ir5	5.6 I _m	-
Ir3	6.6 I _m	-	Ir8	5.6 I _m	-
Ir4	6.6 I _m	-	Ir9	5.6 I _m	-
Ir5	6.6 I _m	-	Ir1	5.7 I _m	2 V _m
Ir6	6.6 I _m	-	Ir7	5.7 I _m	-

The same shading pattern is applied to the Su Do Ku configured array shown in Fig. 5(b) and the current in each row is determined and shown in Table III. It is clear that in the TCT arrangement, rows 8 and 9 are bypassed to extract the maximum power from the array. However, in Su Do Ku arrangement the maximum power occur corresponding to the nominal voltage of the array. The PV characteristics corresponding to this array is shown in Fig. 5(d). The maximum power generated by the array in the TCT arrangement is 3244 W and the GP occurs at 125.8 V. After rearranging the modules according to Su Do Ku puzzle pattern, the power generated by the array is increased by 14.3%.

Case 3 (Short Narrow):

The PV array is subjected to three different insolation levels namely 900 W/m², 600 W/m², and 400 W/m² as shown in Fig. 6(a). The same shading pattern is applied to the Su Do Ku

TABLE IV
LOCATION OF GP IN TCT AND SU DO KU ARRANGEMENT -CASE 3

TCT arrangement			Su Duko puzzle pattern		
Row current in the order in which panels are bypassed	voltage V_a	power P_a	Row current in the order in which panels are bypassed	voltage V_a	power P_a
Ir8	6.1 I _m	9V _m	Ir5	6.8 I _m	9 V _m
Ir9	6.1 I _m	-	Ir9	7.1 I _m	8 V _m
Ir6	7.3 I _m	7V _m	Ir8	7.3 I _m	7 V _m
Ir7	7.3 I _m	-	Ir2	7.5 I _m	6 V _m
Ir5	8.1 I _m	5 V _m	Ir3	7.5 I _m	-
Ir4	8.1 I _m	-	Ir4	7.5 I _m	-
Ir3	8.1 I _m	-	Ir6	7.8 I _m	3 V _m
Ir2	8.1 I _m	-	Ir7	7.8 I _m	-
Ir1	8.1 I _m	-	Ir1	7.8 I _m	-

configured array shown in Fig. 6(b). Table IV depicts the power generated by the array in both TCT and Su Do Ku arrangement. From the PV characteristics corresponding to this array shown in Fig. 6(d), it can be observed that applying the same shading pattern to the Su Do Ku arrangement, increases the generated PV power from 4711 to 5045 W which is 6.6% higher compared to the TCT arrangement. This is due to the dispersion of shaded modules to other rows in the Su Do Ku arrangement [shown in Fig. 6(c)].

Case 4 (Long Narrow):

In this case the PV array is subjected to four different insolation levels namely 900 W/m², 700 W/m², 400 W/m², and 300 W/m² as shown in Fig. 7(a) and (b). The current generated by the array and the corresponding in both TCT and Su Do Ku arrangement are shown in Table V. The GP of both TCT and Su Do Ku arrangement occur at the nominal voltage of the array.

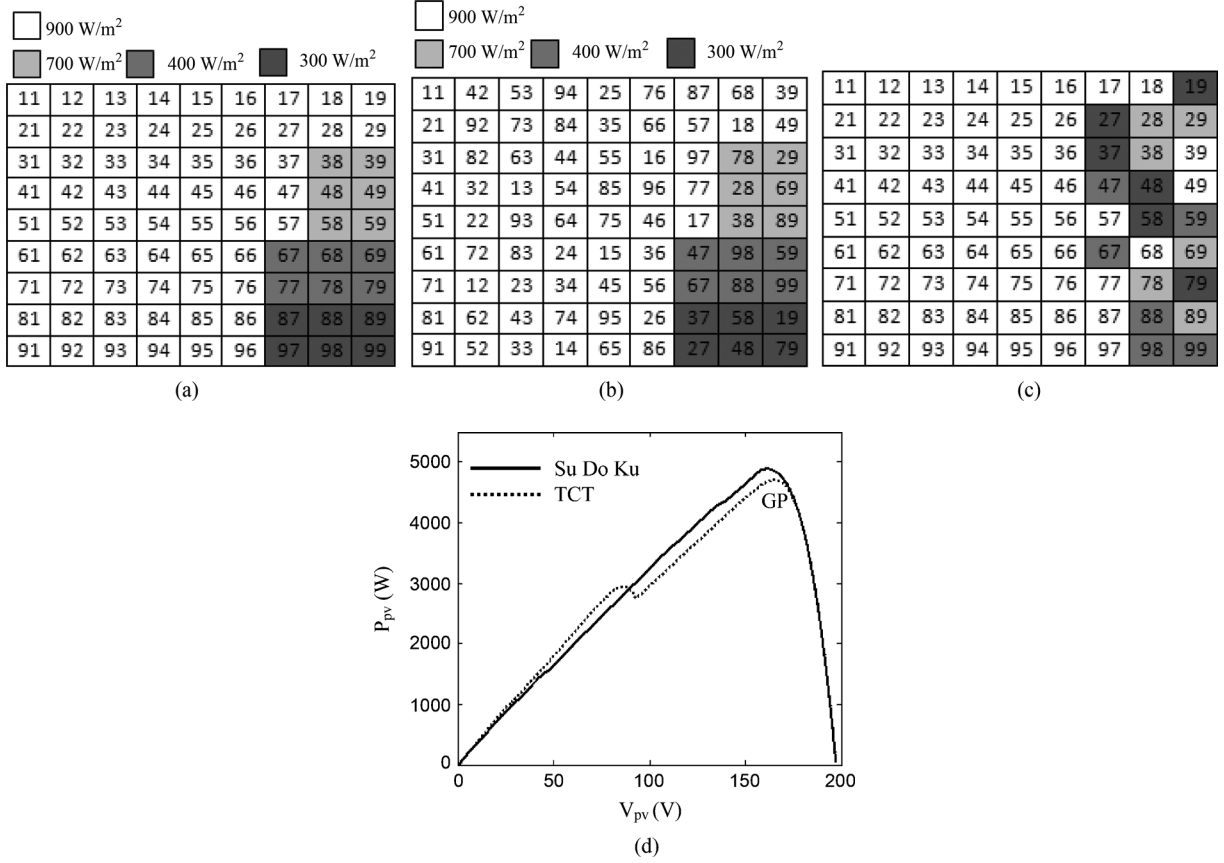


Fig. 7. Shading pattern for case 4. (a) TCT arrangement. (b) Su Do Ku arrangement. (c) Shade dispersion with Su Do Ku arrangement. (d) PV characteristics of the array for TCT and Su Do Ku arrangement for case 4.

TABLE V
LOCATION OF GP IN TCT AND SU DO KU ARRANGEMENT -CASE 4

TCT arrangement				Su Do Ku puzzle pattern			
Row current in the order in which panels are bypassed	voltage V_a	power P_a		Row current in the order in which panels are bypassed	voltage V_a	power P_a	
I_{R8}	$6.3 I_m$	$9 V_m$	$56.7 V_m I_m$	I_{R4}	$7.0 I_m$	$9 V_m$	$63.0 V_m I_m$
I_{R9}	$6.3 I_m$	-	-	I_{R5}	$7.0 I_m$	-	-
I_{R6}	$6.6 I_m$	$7 V_m$	$46.2 V_m I_m$	I_{R2}	$7.1 I_m$	$7 V_m$	$49.7 V_m I_m$
I_{R7}	$6.6 I_m$	-	-	I_{R9}	$7.1 I_m$	-	-
I_{R3}	$7.7 I_m$	$5 V_m$	$38.5 V_m I_m$	I_{R3}	$7.3 I_m$	$5 V_m$	$36.5 V_m I_m$
I_{R4}	$7.7 I_m$	-	-	I_{R7}	$7.3 I_m$	-	-
I_{R5}	$7.7 I_m$	-	-	I_{R6}	$7.4 I_m$	$3 V_m$	$22.2 V_m I_m$
I_{R1}	$8.1 I_m$	$2 V_m$	$16.2 V_m I_m$	I_{R8}	$7.4 I_m$	-	-
I_{R2}	$8.1 I_m$	-	-	I_{R1}	$7.5 I_m$	V_m	$7.5 V_m I_m$

TABLE VI
SUMMARY OF POWER GENERATED IN TCT AND SU DO KU CONFIGURED
STRUCTURE

Case	Maximum power (W)		power enhancement (%)
	TCT structure	Su Do Ku configured structure	
1	3348	4532	26.1
2	3244	4083	20.5
3	4711	5045	6.6
4	4703	4879	3.6

But the power generated by the proposed method is 3.6% higher than the TCT arrangement.

Thus arranging the modules according to “Su Do Ku” puzzle pattern, increases the power produced from the array. As seen from Table VI, in all the four cases the power generated by the Su Do Ku arrangement is higher compared to TCT arrangement.

It is noted that when most of the modules in a row are shaded or under wide shading conditions (case 1 and case 2), the gain obtained in the Su Do Ku arrangement is higher. Moreover, in these cases, the GP which occurs at a reduced voltage in the TCT arrangement is shifted to the MPP voltage of the array in Su Do Ku arrangement.

IV. CONCLUSION

This paper proposes Su Do Ku configuration as a onetime arrangement for the PV modules in an array and demonstrates the improvement in PV power generation under partial shaded conditions. In this approach, the physical location of the modules in the PV array is according to a Su Do Ku puzzle pattern without altering the electrical connection. The Su Do Ku configured structure distributes the effect of shading over the array and reduces the predominant shading of modules in any one row thereby increasing the generated PV power. The performance of the system is studied for different shading patterns and it is revealed that positioning the modules in the array according to Su Do Ku puzzle pattern shows better performance under partially shaded conditions.

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